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# DodgeUI: Leveraging Adaptive Window Movement as Implicit Environmental Cues in Mobile AR

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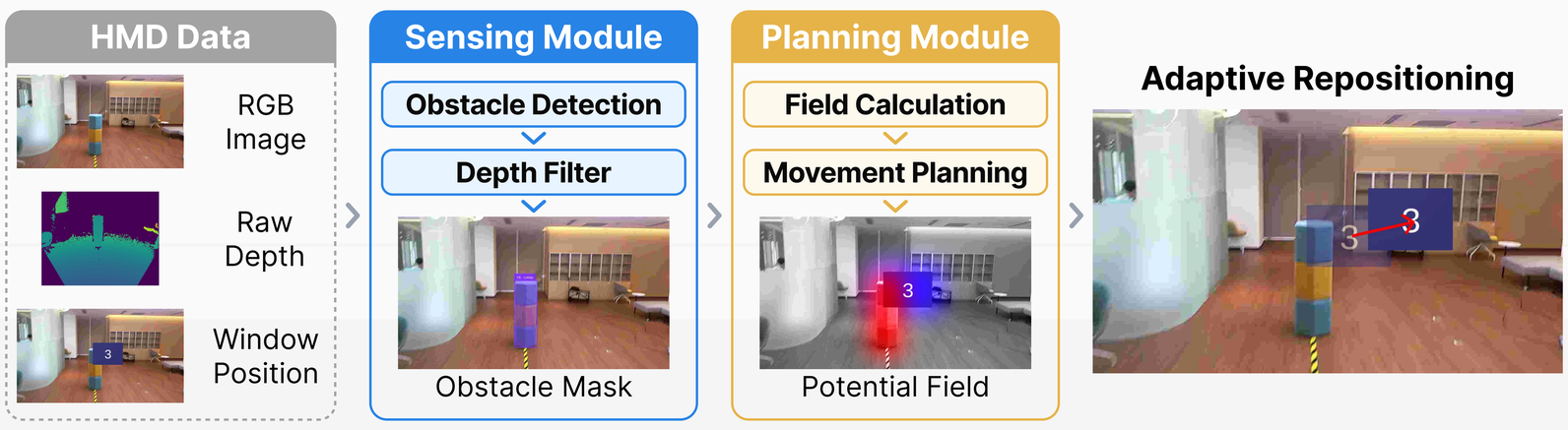
## Motivation

- **Attentional Conflict:** Interacting with AR while walking creates a "dual-task" scenario, splitting limited attentional resources between digital content and physical safety [5, 6, 7].
- **Safety Risks:** Reduced situational awareness may lead to collisions with various obstacles.
- **Visual Clutter:** Existing methods using explicit visual signals like icons or flashes may interfere visual perception [1].

**DodgeUI** - an adaptive interface that dodges physical obstacle

- leverage *Artificial Potential Field (APF)* for dynamic and timely movement calculation and planning
- use window movement as **implicit environmental cues**

## Technical Approach




**Obtain data from Hololens 2**

**Offload computation on Server**

### Artificial Potential Field

**Repulsive Force**
Generate from obstacle

**Attractive Force**
Generate from anchored position

### Adaptive Behaviors of Virtual Window

- **Remain unchanged** with no hazards
- **Slightly shift** with near obstacles
- **Pushed out of view** with close obstacles

## Methodological Approach

### 3×3×2 Within Subject Design

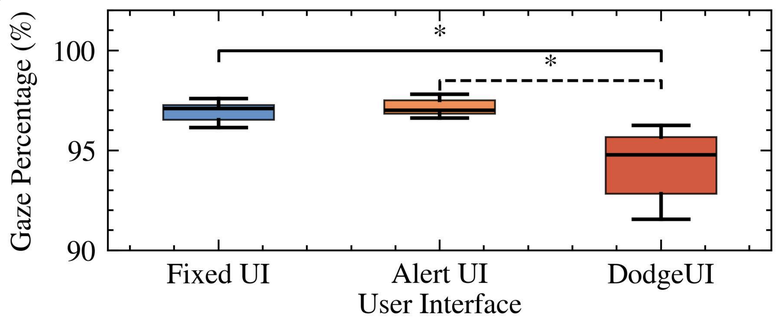
| | | | |
|---|---|---|---|
| User Interface | Fixed | Alert | DodgeUI |
| Distance to Obstacle | Far 1.0m | Near 0.5m | Close 0.0m |
| Obstacle Type | Static Obstacle | Dynamic Pedestrian | |

**24** Participants
14 male, 10 female; age: 22.8±3.5 years

### Continuous Performance Task

Respond to specific numbers on AR display
Click a physical button to respond

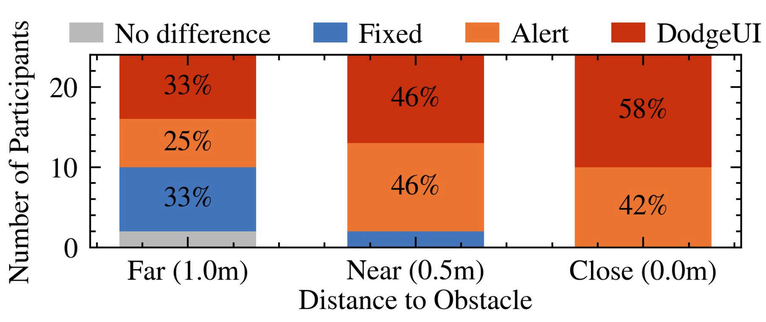
## Results & Discussion




**Eye Gaze** fixation time on the virtual window

**Significant reduction** for DogdeUI compared to other methods.

Window's movement successfully **prompted a shift in visual focus** from the virtual content to the physical environment during potential hazard.




**User Preference**

**Shifted** to DogdeUI in high-risk scenarios.

**Prioritize safety** in complex environment and **perserve interaction** with virtual elements in mild conditions.

**NASA TLX & SUS Scores**
Remain consistent across conditions.

Window movement did not introduce additional task load and maintained ease of use.

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